SUMMARY TABLE: ESSENTIAL MATHEMATICS GRADE 10

Sub-Strand 3.1: Statistics 1 — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 3.1: Statistics 1
Driving Question	DRIVING QUESTION: How can we turn raw data from our school and community into statistics and graphs that help Mama Wanjiku — and other Kenyan decision-makers — solve real problems?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **What Is the Data Telling Us? — Launching the Canteen Queue Puzzle**	Students observed a messy, unsorted list of raw canteen sales data (e.g., number of plates of ugali, githeri, and mandazi sold each day) and immediately noticed it was difficult to read any pattern. They then watched the teacher model converting the same data into a tally-mark frequency distribution table on the board, making patterns — such as which meal is ordered most — instantly visible.	Raw data is essentially invisible information: individual numbers or entries listed without organisation reveal very little on their own. A frequency distribution table, built step-by-step using tally marks (gate tallying: IIII), transforms that chaos into a structured summary that shows each data value and how often it occurs (its frequency). Teachers should note: the key conceptual leap here is that organising data does NOT change the data — it only makes the hidden pattern visible.	Students explained how Mama Wanjiku could use a frequency distribution table built from her canteen sales records to immediately see which meal sells most and which sells least, allowing her to order the right quantities of maize flour, beans, and vegetables without guessing. Pedagogical note: push students to articulate WHY the table helps — the answer should reference 'pattern' and 'frequency', not just 'it looks neater'.
Lesson 2 **From Chaos to Clarity: Collecting Raw Data to Help Mama Wanjiku**	Students conducted a live data-collection exercise, surveying classmates about their favourite school meal (ugali, githeri, chapati, mandazi, or other). Each student recorded responses on a personal tally sheet in real time, experiencing firsthand how raw data accumulates as an unordered list of individual entries before any organisation takes place.	Ungrouped (raw) data is a list of individual data entries that has not yet been sorted or counted. A tally sheet organises data collection efficiently using tally marks so that counting errors are minimised and the transition to a frequency table is straightforward. Teachers should note: a common misconception is that a tally sheet IS the final analysis — clarify that tallying is still a data-collection tool, not yet a summary statistic.	Students explained how using a tally sheet during data collection — rather than writing every answer as a word — saved time, reduced errors, and made it easy to convert their class survey into a frequency table that Mama Wanjiku could use to plan her canteen menu for the week. Pedagogical note: cold-call at least three students to read their tally totals aloud and compare; use discrepancies to discuss the importance of systematic recording.
Lesson 3	Students were presented with a completed	A frequency distribution table can be read in	Students explained how Mama Wanjiku, or

Reading and Interpreting Frequency Distribution Tables	frequency distribution table showing the number of students arriving at school during different 15-minute time intervals in the morning at a Nairobi school. They were asked to read values directly from the table, identify the most and least common arrival times, and calculate the total number of students represented.	two directions: horizontally (to find the frequency of a specific value or category) and vertically (to compare frequencies across all categories or to sum them for the total). The row or column with the highest frequency immediately identifies the modal class. Teachers should note: students often confuse the 'value/category' column with the 'frequency' column — use colour coding or underlining to distinguish the two during instruction.	the school deputy principal, could read a completed frequency distribution table of late arrivals to identify the peak congestion window and decide whether to open the school gate earlier — directly connecting table-reading skills to a real community decision. Pedagogical note: ask students 'What decision would YOU make, and what number in the table tells you that?' to anchor interpretation in action.
Lesson 4 **From Chaos to Clarity: Building Frequency Distribution Tables**	Students were given a set of 30 raw data values representing the number of mandazi sold by Mama Wanjiku on each of 30 school days. They worked in pairs to sort the data, identify unique values, apply tally marks, and complete a blank frequency distribution table, recording each value and its corresponding frequency.	A frequency distribution table organises ungrouped raw data by recording each possible data value and its frequency (how often it occurs). The construction process follows four steps: (1) identify all unique values, (2) list them in ascending order, (3) tally each occurrence, (4) write the final frequency count. The sum of all frequencies must equal the total number of data entries — this is the built-in check for accuracy. Teachers should note: insist students always verify their frequency total against the raw data count before moving on.	Students explained that by building a frequency distribution table from Mama Wanjiku's 30-day sales record, she can instantly see her most and least common daily sales figures, helping her set a realistic daily production target — turning 30 scattered numbers into one actionable table. Pedagogical note: use the 'frequency total check' as a non-negotiable classroom norm from this lesson forward.
Lesson 5 **Visualising Data: Drawing Bar Graphs and Pictographs**	Students examined two visual representations of the same canteen data: a hand-drawn bar graph and a pictograph (using a maize-cob symbol to represent 5 meals). They compared the two displays, noting which was faster to read, which required a key, and which made the difference between categories most visually striking.	Bar graphs use rectangular bars whose height (or length) is proportional to frequency, with a labelled x-axis (categories) and y-axis (frequency). Pictographs use a repeated symbol with a key indicating the value of each symbol; partial symbols represent fractional values. Both are valid ways to display categorical or discrete data visually, but bar graphs allow more precise reading of exact values while pictographs are more immediately engaging for a general audience. Teachers should note: a common error is drawing bars of unequal width — emphasise uniform bar width as a rule.	Students explained how Mama Wanjiku could display her weekly sales data as a bar graph on a notice board outside her canteen so that students, teachers, and parents could see at a glance which meals are most popular — making the data accessible to community members who may not read number tables easily. Pedagogical note: ask 'Who is your audience?' to drive home that graph choice depends on the reader, not just the data.
Lesson 6 **Measures of Central Tendency: Mean and Mode — Finding the 'Typical' Meal for Mama Wanjiku**	Students were given Mama Wanjiku's sales data for five days: 55, 60, 55, 60, 60 meals served. They first predicted which single number best represents a 'typical' day, then calculated the mean (290 ÷ 5 = 58) and identified the mode (60, appearing three	The MEAN is calculated by summing all data values and dividing by the number of values (sum ÷ n). The MODE is the value that appears most frequently in the data set; a data set can have one mode (unimodal), two modes (bimodal), or no mode. Both are	Students explained that for Mama Wanjiku, the MODE (60) is more actionable than the mean (58) because it tells her the most common number of meals she actually serves on a single day, helping her set a reliable daily cooking target — while the

	times). They observed that these two values are different and discussed which one feels more 'representative'.	measures of central tendency — numerical summaries that describe the 'centre' or 'typical value' of a data set — but they answer slightly different questions: the mean balances all values, while the mode identifies the most common value. Teachers should note: the mean is sensitive to extreme values (outliers); the mode is not — this distinction becomes critical in Lesson 7.	mean gives a useful average for weekly planning. Pedagogical note: use cold-calling to ask at least four students 'Which measure would YOU use if you were Mama Wanjiku, and why?' — accept both answers if reasoning is sound.
Lesson 7 **The Middle Value: Median — Finding the Centre of Mama Wanjiku's Data**	Students physically lined up in order of the number of mandazi they claimed to eat per week (a quick kinaesthetic data set), then identified the student standing exactly in the middle. They then applied the same logic to written data sets — first an odd-numbered set (finding the exact middle value) and then an even-numbered set (averaging the two middle values) — to derive the median in each case.	The MEDIAN is the middle value of a data set that has been arranged in ascending (or descending) order. For an ODD number of values, the median is the single middle value at position (n+1)/2. For an EVEN number of values, the median is the mean of the two middle values at positions n/2 and (n/2)+1. The median is a measure of central tendency that is NOT affected by extreme outliers, making it especially useful when data contains very high or very low values. Teachers should note: the most common procedural error is forgetting to sort the data first — make sorting a mandatory, visible step before any median calculation.	Students explained that if Mama Wanjiku's daily sales data for one week included one unusually low day (e.g., a public holiday), the median would give her a more reliable picture of her 'typical' trading day than the mean, because the outlier drags the mean down without affecting the median. Pedagogical note: present a data set with a clear outlier and ask students to calculate both mean and median, then decide which Mama Wanjiku should report to the school board — this surfaces the practical power of choosing the right measure.
Lesson 8 **Putting It All Together: Statistics in Action for Kenyan Decision-Makers**	Students reviewed a complete, realistic data scenario: Mama Wanjiku's canteen sales data for one full school month (20 days), presented as raw data. They were tasked with building the frequency distribution table, drawing a bar graph, and calculating mean, mode, and median — applying every skill from Lessons 1–7 in sequence as a culminating performance task.	All three measures of central tendency (mean, mode, median) can be calculated from the same data set and each reveals a different aspect of the data's centre. Together with a frequency distribution table and a bar graph, they form a complete statistical toolkit for summarising and communicating data. The choice of which statistic to report depends on the purpose and audience: mean for balanced overall averages, mode for the most common value, median when outliers are present. Teachers should note: this lesson is the integration point — resist the urge to re-teach individual skills; instead, coach students on the decision-making process of choosing the right tool for the right question.	Students explained — in a written summary addressed directly to Mama Wanjiku — which statistic best supports each of her specific decisions (daily cooking target, weekly purchasing, reporting to the school board), demonstrating that statistics are not abstract calculations but practical tools that real Kenyan community members can use to make better, evidence-based decisions. Pedagogical note: the written summary doubles as a formative assessment; look for students who correctly match the statistic to the decision context, not just those who calculate correctly.